

CE301.01 Parallel Computing

Date: 20.11.2013, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

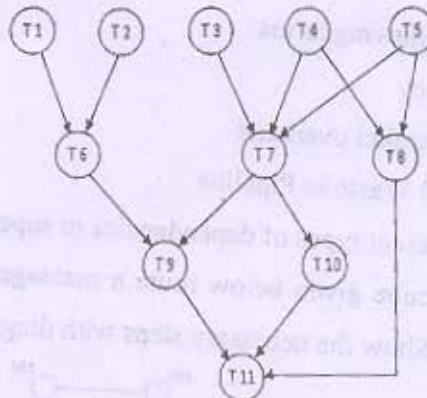
SECTION - I

Q - 1 (a) What is task dependency graph? "Task-dependency graphs can be disconnected and the edge-set of task-dependency graph can be empty". Justify the statement with suitable example. [04]

(b) Explain granularity of the decomposition. How degree of concurrency is related to granularity. [03]

Q - 2 (a) For the task graph given below, determine the following: [06]

1. Maximum degree of concurrency
2. Critical path length
3. The maximum achievable speed if the number of process is limited to (a) 2, (b) 4, and (c) 6



(b) What is recursive decomposition and exploratory decomposition? Explain with suitable example. [04]

OR

(b) What is data decomposition? Explain the owner-computes rule. [04]

(c) List and explain the characteristics of tasks. [04]

OR

(c) What is parallel algorithm model? Write the difference between work pool model and master-slave model. [04]

Q - 3 (a) Compare and contrast the blocking message passing operations and non-blocking message passing operations. [04]

(b) Explain the functions for sending and receiving messages in MPI. [06]

OR

(b) Explain the situation with suitable example which may lead to deadlock in MPI program. Explain the use of MPI_Status structure in MPI. [06]

(c) What is thread? Write the advantages of thread programming over message-passing programming model. [04]

OR

(c) Explain the functions used to create and terminate the thread. [04]

SECTION - II

Q - 4(a) Define parallel computing. Give its application and limitations. [04]

(b) A hypercube contains 32 nodes. Find its diameter, bisection width and arc connectivity. [03]

Q - 5 (a) How to embed a linear array into a hypercube? Elaborate in detail. [07]

(b) Explain one to all broadcast on a hypercube. [07]

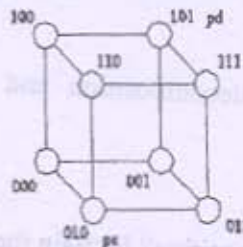
OR

(b) 1. Define the following terms: [03]

- Efficiency
- Total Parallel overhead
- Vertical Waste in Pipeline

2. Explain different types of dependencies in superscalar execution. [04]

Q - 6 (a) For the hypercube given below route a message from p_s (010) to p_d (101) using e-cube routing. Show the necessary steps with diagram. [07]



(b) Describe a circular 5-shift operation in a 4 x 4 mesh. [07]

OR

(b) Explain omega network with its advantages and disadvantages. [07]
